Tracing biological priors in cancer drug response prediction:

A narrative review and diagnostic GDGN case study

Abstract

Biological knowledge enters cancer drug response models through graph connectivity, auxiliary learning objectives and conditional attention. Establishing how each route contributes to a prediction is central to designing interpretable models. This narrative review develops a diagnostic framework connecting claims about attention-based explanation, pretraining transfer and topological benefit to their corresponding comparisons. A retrospective GDGN case study gives the framework a concrete empirical basis. At fixed checkpoints, replacing protein-association inputs with self-loops changes Pearson correlation (PCC) by -0.0007 in the A4 attention encoder and -0.1458 in the B4 dual encoder, revealing sharply different relational sensitivities. Across ten B4 interpretation cases, attention and conditional-omics Integrated Gradients have a mean rank correlation of -4.43×10^{-4}; attention top-20 sets show extensive reuse across cases, with mean Jaccard overlap 0.9192, compared with 0.0250 for attribution sets. Annotation reconstruction further shows how feature selection changes the prior available to a model. These findings establish architecture-specific relational dependence and distinct readout and attribution profiles in the examined model family. The framework links these observations to matched transfer and topology comparisons, providing a coherent way to evaluate what biological priors contribute and how that contribution can be explained.

Keywords: cancer drug response; biological priors; graph neural networks; cross-attention; Integrated Gradients; transfer learning; diagnostic evaluation

1 Introduction

Biological priors connect molecular measurements with the processes through which drugs act on cancer cells. A central evaluation question is how those priors participate in a model's predictions. We address this question through a diagnostic framework linking readout, initialization and topology to the comparisons that reveal their contributions. The framework combines a narrative literature synthesis with a retrospective GDGN case study. Its clearest empirical illustration is the sharply different response of the A4 and B4 checkpoints to protein–protein interaction (PPI) input replacement by self-loops, with PCC changes of -0.0007 and -0.1458, respectively.

Pharmacogenomic resources such as the Genomics of Drug Sensitivity in Cancer (GDSC) and the Cancer Cell Line Encyclopedia (CCLE) provide the basis for response modelling by linking cellular profiles with measured sensitivity[1–3]. For a drug d and cell line c, a predictor estimates response y_{cd} from chemical and cellular inputs. DeepCDR couples chemical graphs to multi-omics representations, MOLI integrates cellular modalities, and DrugCell organizes cellular representations through biological systems[4–6]. Protein-association networks, drug–gene annotations and attention mechanisms introduce further routes for combining drug identity with cellular context.

We organize this question around three routes: *readout*, *initialization* and *topology*. Attention weights describe the selection of contextual representations; input attribution describes a specified output difference. Pretraining shapes the starting representation; its downstream contribution is measured with representation access held constant. Biological topology organizes message passing; its predictive contribution is measured against an appropriate retrained structural control. Connecting each route to a defined comparison gives a shared vocabulary to studies that otherwise use different explanatory objects and experimental protocols.

B4 combines substantial graph sensitivity with nearly zero attention–Integrated Gradients (IG) rank agreement in the selected interpretation cases. These observations place computational dependence and explanatory readout alongside predictive accuracy. Their joint analysis leads to the central insight of this review: the

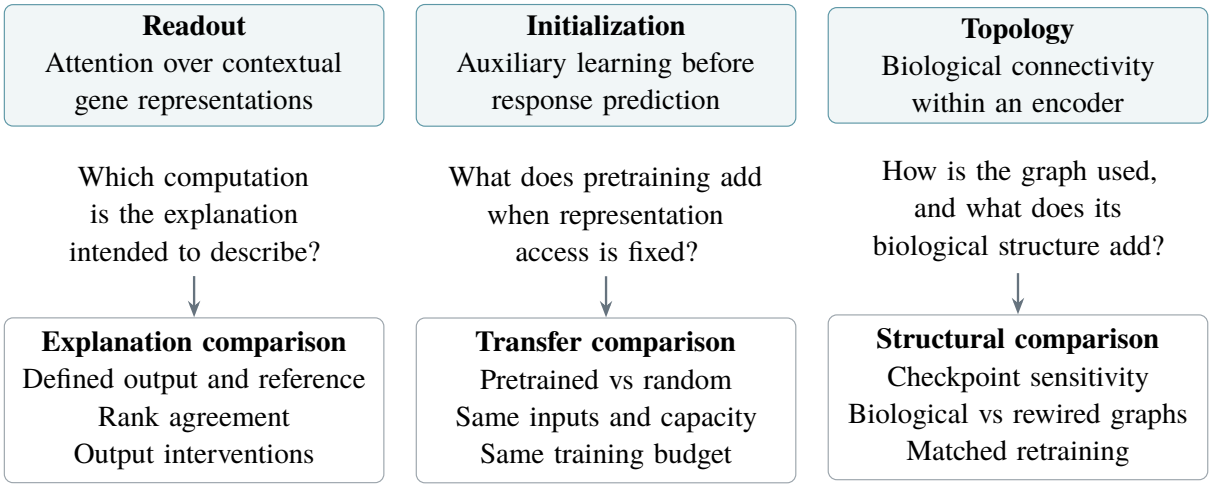


Figure 1: A diagnostic framework for biological priors. Each information route is linked to a scientific question and a corresponding comparison. Readout analysis connects attention to a declared output function; transfer analysis holds representation access constant; topology analysis pairs checkpoint sensitivity with matched retraining. Prediction task, feature coverage and model capacity define the common evaluation context.

role of a biological prior is identified by the information route and experimental contrast together. Figure 1 summarizes the framework.

2 Biological priors as information routes

2.1 Chemical structure, cellular relations and functional organization

Graph-based drug response models represent several kinds of knowledge. Chemical graphs describe atoms and bonds. DeepCDR and GraphDRP learn molecular representations from these graphs and combine them with cellular features[4, 7]. MOLI emphasizes modality-specific cellular encoding and fusion for binary response classification[5]. Together, these designs illustrate how chemical structure and cellular measurements supply complementary information about response.

Biological relation graphs provide cellular context through associations among gene products and links between compounds and genes. Functional hierarchies supply another form of organization: DrugCell structures cellular representations according to biological systems[6]. DRPreter uses network-informed pathway representations with a transformer-based predictor[8]. The represented prior determines the scientific comparison. A chemical encoder is evaluated against alternative molecular representations, while a cellular relation graph calls for controls over biological connectivity, node features and encoder capacity.

Response-derived graphs have a further role. GraphCDR represents drug–cell responses as relations and incorporates contrastive regularization into response classification[9]. Its relational signal is tied to supervised response observations, whereas a protein-association graph encodes external biological knowledge. Table 1 locates representative methods within this landscape.

2.2 Readout and parallel paths

Attention creates a conditional interface between modalities. In PaccMann, gene-expression context conditions attention over chemical sequence features[10]. In GDGN, a drug query reads cellular gene representations. These directions of conditioning specify what the weights summarize. In a graph encoder, each

Table 1: Representative modelling strategies, grouped by the information they organize. Their relevance is expressed through representation and evaluation questions.

Approach	Representation or integration	Scientific emphasis
DeepCDR[4]	Drug molecular graph and multi-omics cellular inputs	Chemical structure combined with cellular integration
GraphDRP[7]	Drug molecular graphs and genomic cellular features	Molecular encoding and predictive readout
MOLI[5]	Multi-omics fusion for binary response classification	Complementary cellular modalities
PaccMann[10]	Expression-conditioned attention over chemical sequence features	Direction of conditioning in multimodal attention
GraphCDR[9]	Drug–cell response graph and joint contrastive regularization	Response relations and task-related auxiliary learning
DrugCell[6]	Biologically organized cellular hierarchy	Functional organization of cellular representations
DRPreter[8]	Network-informed pathway representations and a transformer	Biological structure with conditional prediction

value vector already incorporates relational context, so the readout operates on a learned representation of the cellular state.

Parallel encoders expand the routes available to a predictor. A graph branch provides relationally contextualized features; a flatten branch provides direct access to molecular measurements. Their combination allows the final predictor to use both representations. This architecture also makes branch-specific analysis valuable: a weight within the graph readout and an attribution of the complete predictor describe different levels of the same computation.

2.3 Coverage defines the available prior

Biological knowledge reaches a model through identifier mapping, annotation filtering and intersection with the measured feature universe. Each step determines which relations remain available for a particular drug. STRING integrates physical interactions and functional protein associations, while DGIdb aggregates drug–gene interaction annotations[11, 12]. Their semantics guide the interpretation of the model relations derived from them.

Coverage is therefore a property of the assembled model input. A useful description reports mapped drugs, drugs with represented annotations, unique retained relations and targets per drug. This connects data assembly to model behaviour and makes subgroup comparisons biologically interpretable. The GDGN coverage reconstruction below illustrates the impact of the gene universe on this description.

3 A framework that links claims to comparisons

3.1 Readout: align the object of explanation

The attention literature provides a basis for specifying explanatory claims. Jain and Wallace examined agreement between attention and feature-importance measures[13]; Wiegreffe and Pinter emphasized explanation definitions and model-aware controls[14]. Clinical-note research likewise separated predictive utility from explanatory support[15]. Their shared implication for drug response modelling is to declare the computational object that an explanation is intended to describe.

A simplified readout makes this object explicit. Let $z = \sum_i a_i v_i$, with input-dependent attention weights and contextual values. Differentiation gives

$$\frac{\partial z}{\partial x_j} = \sum_i \left(a_i \frac{\partial v_i}{\partial x_j} + v_i \frac{\partial a_i}{\partial x_j} \right). \tag{1}$$

Graph propagation allows a value vector to depend on measurements at other genes. Attention weights capture one part of this computation; input derivatives include changes in both values and weights. Their ranking agreement is thus an empirical signature of the model and its explanation target.

IG allocates a prediction difference along a path from reference x' to input x[16]:

$$\mathrm{IG}_j(x; x') = (x_j - x'_j) \int_0^1 \frac{\partial f(x' + t(x - x'))}{\partial x_j} \, \mathrm{d}t. \tag{2}$$

For a multi-input predictor, defining this path also specifies how the other inputs are treated. GDGN's historical implementation attributes the predictor $F_{d,c}(x) = f(x, p_c, z_d)$ to omics while holding observed pathway features p_c fixed. Its gene-level ranking uses the four-channel magnitude

$$s_i = \left(\sum_{k=1}^{4} \mathrm{IG}_{ik}(x_c; x'_c \mid p_c)^2 \right)^{1/2}. \tag{3}$$

Signed channel attributions allocate $F_{d,c}(x_c) - F_{d,c}(x'_c)$ under the IG completeness property; the L2 magnitude supplies a ranking. This distinction gives the archived result a precise meaning: conditional-omics attribution of the complete predictor.

A coherent explanation study then connects three forms of evidence. Rank agreement compares summaries; interventions evaluate output dependence; annotation or pathway overlap evaluates biological alignment. Parameter randomization tests whether an attribution tracks learned parameters[17], while attention interventions examine dependence on the selected representations[18]. GNNExplainer and graph-attribution research extend these questions to structured explanatory objects[19, 20]. Together, these comparisons turn an attention map into a testable account of a specified computation.

3.2 Initialization: isolate the transfer contrast

Pretraining shapes the representation from which response learning begins. Graph pretraining studies show that the choice and combination of auxiliary tasks affect transfer[21]. GraphCL contrasts augmented graph views, and GraphMAE reconstructs masked features[22, 23]. Their objectives encourage different forms of information retention. For drug response, the relevant question is how this information supports a cellular state's response to a compound.

GDGN's historical pretraining implementation masks protein-association and drug–gene edges, incorporates cell-specific omics into a heterogeneous encoder, and predicts the masked relations against sampled negatives. The objective combines relation-specific binary cross-entropy terms, with a weighted drug–gene term and an optional margin component. This supplies an initialization for subsequent continuous-response regression. GraphCDR's jointly trained contrastive classification objective provides a complementary example of incorporating an auxiliary signal during response learning[9].

Transfer is identified by comparing pretrained and randomly initialized models with the same downstream inputs. Let $P \in \{0, 1\}$ indicate pretraining and $E \in \{0, 1\}$ indicate access to the main-graph drug representation. For a higher-is-better downstream metric M, the contrast is

$$\Delta_P(E) = M(P = 1, E) - M(P = 0, E). \tag{4}$$

A two-by-two design evaluates this contrast at each representation setting. Matching capacity, optimization budget and initialization of untransferred modules makes the role of pretraining identifiable within that design. Learning curves add information about convergence and final accuracy.

3.3 Topology: distinguish sensitivity from structural benefit

Two comparisons reveal complementary properties of a relation graph. A fixed-checkpoint perturbation measures the trained predictor's dependence on its relational input. Writing θ for the fixed parameters and $\mathcal{G}$ for that input, the metric change is

$$\Delta_{\text{input}} = M(\theta, \mathcal{G}_{\text{perturbed}}) - M(\theta, \mathcal{G}_{\text{original}}). \tag{5}$$

This comparison localizes a computational dependence under a stated intervention. It is particularly useful for comparing how different architectures use the same class of relation.

A retrained topology comparison evaluates the predictive information supplied by biological connectivity:

$$\Delta_{\text{topology}} = M(\text{trained with biological topology}) - M(\text{trained with matched rewired topology}). \tag{6}$$

Degree structure, node features, edge-weight distributions, model capacity and training budget define the matching conditions. For an edge-weighted graph, the protocol also specifies how weights follow the rewiring. Pairing the two contrasts yields a fuller interpretation: the first describes use by a fitted model, and the second evaluates the structural prior under a controlled training comparison.

4 GDGN as a diagnostic case study

4.1 Data, evidence and representation paths

The case study uses GDGN development records through the B4 configuration. Its inputs describe 404 cell lines, 184 drugs, 8,412 genes and 186 pathway features, with continuous LN_IC$_{50}$ response. Predictive comparisons use the archived random-pair test reports, each containing 6,655 drug–cell-line pairs. The narrative synthesis draws on primary method, resource and evaluation papers selected for the three information routes.

The analysis reconstructs statistics from saved evaluation and perturbation reports, interpretation records and annotation files. Spearman correlations are summarized from stored scalars; top-20 overlaps are recalculated from gene lists. Source hashes and machine-readable results accompany the article. Appendix A gives record-level definitions and the computational scope.

GDGN combines a cellular relation graph with a drug molecular encoder and pathway features. Its operational PPI and drug–target interaction (DTI) relations derive from STRING protein associations and DGIdb drug–gene annotations, respectively[11, 12]. These relation names follow the implementation; annotation recovery measures overlap with the retained drug–gene records.

Let $x_c \in \mathbb{R}^{G\times 4}$ contain expression, mutation, copy-number and methylation measurements, p_c denote pathway inputs, and H_c denote graph-encoded gene representations. A4 reads H_c using drug-conditioned attention. B4 combines a residual graph readout with a parallel flatten encoder:

$$c_g = b\left(G^{-1}\sum_{i=1}^{G} H_{c,i}\right) + \lambda\,\text{MHA}(q(z_d), H_c, H_c), \tag{7}$$

$$c_f = \text{FlattenMLP}(x_c, p_c), \tag{8}$$

$$\widehat{y}_{cd} = h([c_g, c_f, z_d, z_p]). \tag{9}$$

Here z_d and z_p are molecular drug and pathway representations, b is a learned projection, and λ is a learned residual scale. The fused input contains 256 graph-cell, 256 flatten-cell, 128 drug and 64 pathway dimensions. Stored attention describes the attention component of c_g; conditional-omics IG describes the complete predictor's differentiable routes from x_c to its output. This architecture makes the distinction between readout and output attribution directly observable.

Table 2: Changes under fixed-checkpoint graph perturbations. Each change is relative to the original relational input of the corresponding model.

Input change	A4 ΔPCC	B4 ΔPCC	A4 ΔRMSE	B4 ΔRMSE
Randomize PPI destinations	-0.000203	-0.082900	+0.001955	+1.180475
Randomize DTI destinations	-0.000403	-0.034092	+0.003315	+0.521284
Randomize both destinations	-0.000182	-0.051378	+0.001373	+0.994495
Replace PPI with self-loops	-0.000700	-0.145801	+0.005621	+1.440684
Replace DTI with placeholder	-0.000893	-0.070219	+0.009808	+1.317164
Apply both replacements	-0.000829	-0.148199	+0.009016	+1.451681

$\Delta M = M_{\text{perturbed}} - M_{\text{original}}$. Original PCC/RMSE: A4, 0.919079/1.153252; B4, 0.935352/1.038872. Destination randomization preserves the source-edge list and edge count. Parameters remain fixed.

4.2 Contrasting relational dependence in A4 and B4

A4 and B4 show sharply different relational sensitivities under fixed-parameter perturbations (Table 2, Figure 2). Replacing PPI inputs with self-loops shifts A4 PCC from 0.919079 to 0.918379 and B4 PCC from 0.935352 to 0.789551. The corresponding changes are -0.000700 and -0.145801. B4 therefore exhibits substantial dependence on the original PPI input under this intervention, while A4 is comparatively insensitive.

The same ordering appears under a different operation. Randomizing PPI destinations gives PCC changes of -0.000203 for A4 and -0.082900 for B4. DTI replacement gives -0.000893 and -0.070219, respectively. For PPI self-loop replacement, the associated RMSE values are 1.158873 in A4 and 2.479556 in B4, relative to their original 1.153252 and 1.038872. Thus both correlation and response-scale error distinguish the checkpoints' relational dependence.

These interventions have explicit operational definitions. PPI replacement uses self-loops; DTI replacement keeps one placeholder edge per direction. Destination randomization preserves the source-edge list and edge count, drawing new destinations uniformly with replacement. Forward and reverse relations are randomized separately. The measured effects describe dependence on these relational inputs, including the degree and reciprocity changes induced by the operations.

The architecture-specific profiles provide the key empirical insight: the presence of a biological graph in a model does not determine how strongly its fitted predictor uses that input. B4's parallel representation paths coexist with pronounced graph sensitivity. The two contrasts in Equations 5 and 6 give this observation its place in the framework: checkpoint dependence provides a focused starting point for evaluating the retained structural prior.

4.3 Readout reuse and input attribution provide different profiles

The ten B4 interpretation records show a mean stored attention–IG Spearman correlation of -0.000442738, with values ranging from -0.008458427 to 0.012438615 (Figure 3). Eight cases share the top-five attention signature LIG3, CASP14, SERPINA1, ABL2 and ACKR3. Across all ten cases, attention produces three distinct top-20 gene sets.

The cross-case comparison quantifies this reuse. Across the 45 pairs formed from the ten cases, mean Jaccard overlap, $J(A, B) = |A \cap B|/|A \cup B|$, is 0.9192 for attention and 0.0250 for conditional-omics IG. Attention therefore selects a highly recurrent set of contextual gene representations, while the attribution rankings vary across the selected contexts. These are descriptive comparisons of shared cases.

The result is informative because the model offers a clear computational explanation target for each profile. Attention weights select values after graph propagation; IG allocates an output difference across molecular inputs and all their differentiable prediction routes, conditional on the pathway input. Equation 1 connects

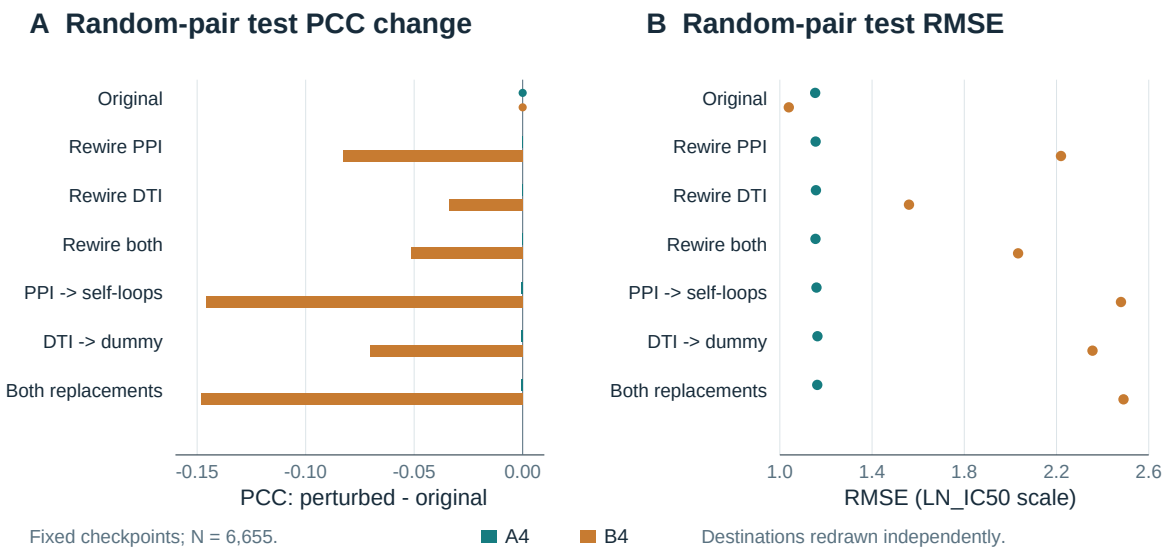


Figure 2: Architecture-specific relational dependence in A4 and B4. The panels show PCC changes relative to each checkpoint's original input and RMSE on the stored response scale. PPI self-loop replacement and destination randomization both expose distinct sensitivity profiles. DTI replacement retains a placeholder relation. Parameters remain fixed for all operations.

Table 3: Ten archived interpretation cases. Relative sensitivity and resistance identify the development examples.

Case	Drug	Cell line	Context	T_d	IG hits	ρ
1	GSK2606414	EOL-1	sensitive	3	0	0.001751
2	GSK2606414	AsPC-1	resistant	3	0	-0.000741
3	LGK974	EOL-1	sensitive	2	0	0.012439
4	LGK974	NMC-G1	resistant	2	0	0.006332
5	Daporinad	RCH-ACV	sensitive	1	0	-0.008458
6	Daporinad	EKVX	resistant	1	0	-0.007368
7	BEN	KYM-1	sensitive	0	N/A	-0.001929
8	BEN	COLO-678	resistant	0	N/A	0.006272
9	Acetalax	HCC2157	sensitive	0	N/A	-0.006216
10	Acetalax	AsPC-1	resistant	0	N/A	-0.006507

T_d: annotated drug-associated genes in the 8,412-gene universe. IG hits: annotation overlap among the top 20 genes. N/A: annotation recovery unavailable because $T_d = 0$. ρ: stored attention–IG Spearman correlation.

the two levels. The observed rank separation identifies the interpretation question that matters for this architecture: how contextual readout and direct molecular dependence combine to produce a response prediction.

Annotation recovery supplies another view of the rankings (Table 3). Six records, representing three drugs, have retained drug-associated genes in the model's annotation universe; four are labelled N/A. The IG top-20 sets contain zero annotation hits in the six evaluable records. With $K = 20$ selected genes, $G = 8{,}412$ represented genes and T_d represented drug-associated genes, uniform sampling has expectation

$$\mathbb{E}[H_d] = \frac{KT_d}{G}. \tag{10}$$

The summed expectation across the six evaluable records is 0.02853. This places the observed overlap in a sparse-annotation setting and makes represented coverage an essential part of the biological interpretation.

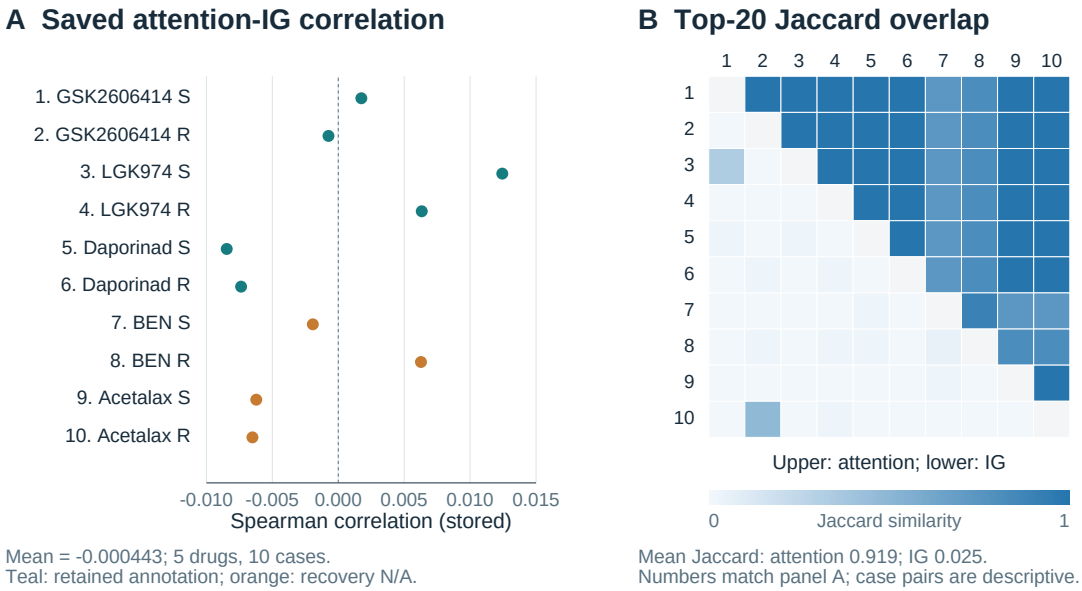


Figure 3: Distinct readout and attribution profiles in ten B4 cases. Stored attention–IG correlations cluster near zero. The heatmap compares cross-case top-20 sets: attention in the upper triangle and IG in the lower triangle. Extensive attention-set reuse contrasts with context-varying attribution sets. Annotation recovery is evaluated in the six cases with represented drug–gene annotations.

4.4 Transfer and coverage depend on the information available

The A1/A4 comparison illustrates the value of a matched transfer contrast. A1 combines pretraining with access to the main-graph drug embedding; A4 combines random initialization with that embedding disabled. Enabling the embedding expands the fused input from 448 to 576 dimensions and adds 65,664 regression-head parameters. Their recorded PCC values, 0.9162 and 0.9191, describe these composite configurations. Equation 4 organizes the corresponding transfer question by holding representation access and head capacity fixed.

Coverage reconstruction shows how information access changes earlier in the pipeline. Seven of 184 mapped drugs have no drug-associated genes in the filtered annotation records. After intersection with the 8,412-gene model universe, eleven drugs have zero represented annotations: four additional drugs have annotations outside that universe. The intersection yields 1,124 unique drug–gene pairs, matching the relation count reported in the perturbation records.

This change gives a concrete meaning to coverage at the model level. Database annotation, successful mapping and representation in the gene universe are distinct stages. Reporting them separately identifies the prior available to each prediction and supports interpretable comparisons across drugs. The same principle connects transfer and coverage: scientific interpretation follows the information actually supplied to the predictor.

4.5 Prediction anchors the diagnostic comparison

The archived random-pair results place these diagnostic observations within a predictive setting (Table 4, Figure 4). B4 records PCC 0.9354 and RMSE 1.0389, compared with A4's 0.9191 and 1.1533. The flatten comparator records PCC 0.9378 and RMSE 1.0191; the project-adapted DeepCDR comparator records 0.9215 and 1.1511.

Historical descriptions place A4 at approximately 1.77 million parameters and B4 at 19.3 million. The configuration comparison therefore concerns complete predictive systems, with changes in both representation paths and capacity. The diagnostic value of the case study lies in relating their predictions to how

Table 4: Archived random-pair test results. Each row represents one historical run with 6,655 test pairs and recorded seed 42.

Model	Setting	PCC	Spearman	RMSE	MAE
A0	Attention; pretrained, slow encoder updates	0.9055	0.8740	1.2587	0.9138
A1	Attention; pretrained, main-graph drug access	0.9162	0.8832	1.1925	0.8882
A4	Attention; random initialization	0.9191	0.8861	1.1533	0.8564
B1	Residual graph readout	0.9232	0.8894	1.1278	0.8387
B3	Concatenated graph readout	0.9175	0.8841	1.1760	0.8649
B4	Residual graph and flatten encoders	0.9354	0.9068	1.0389	0.7735
Flatten	Flattened cellular inputs	0.9378	0.9101	1.0191	0.7549
DeepCDR (adapted)	Project-adapted DeepCDR	0.9215	0.8893	1.1511	0.8396

Configurations differ in architecture, initialization, representation access and optimization. These values provide descriptive predictive context. The DeepCDR row is the project's adapted comparator.

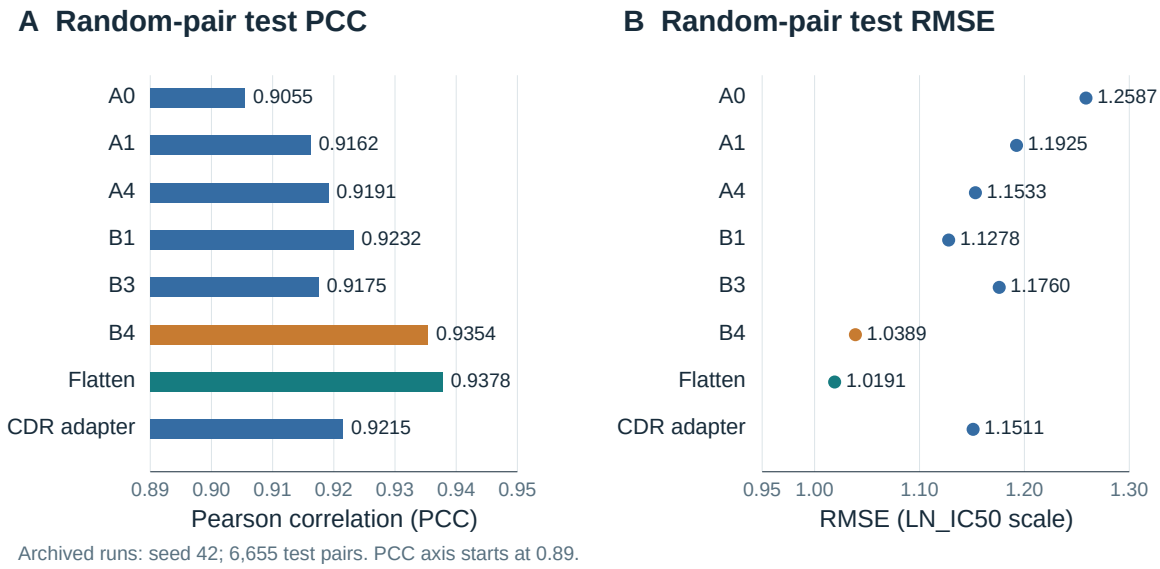


Figure 4: Predictive context for the GDGN diagnostic case study. Archived PCC and RMSE summarize individual runs on random-pair test reports containing 6,655 observations each. The configurations span attention, residual, parallel and flatten cellular representations; DeepCDR denotes the project's adapted comparator.

they use relational and molecular information. Predictive metrics, attention profiles and graph sensitivity together describe this relationship.

5 Designing coherent evaluations

The framework makes the intended prediction task part of every comparison. Random-pair evaluation tests new combinations among drugs and cell lines that can already appear during training. Drug-held-out and cell-line-held-out evaluation test transfer across the corresponding entity boundary. DrEval provides a standardized treatment of these settings and of drug and cell-line mean effects[24]. Pooled metrics, within-drug metrics and training-only mean or additive baselines provide complementary descriptions of response prediction.

The available external knowledge also defines the task. A drug with no measured training responses may have a known molecular structure and curated targets. Retaining those priors defines an informed unseen-response setting. A protocol that removes test-drug relations evaluates a different information boundary. Stating this boundary links the benchmark to the setting in which the model will be used.

Within a declared task, a compact evaluation can follow the three information routes. Readout analysis specifies the output function and reference, checks attribution numerics, compares rankings and intervenes on selected representations. Transfer analysis independently varies pretraining and representation access. Topology analysis pairs fixed-checkpoint changes with biological, matched rewired and relation-suppressed training controls. Each comparison has a defined inferential role.

Uncertainty is attached to the same role. Training seeds describe variation in fitted models, alternative rewired graphs describe variation in a structural control, and resampling fixed predictions describes sampling uncertainty. Shared drugs and cell lines induce dependence among response observations. Preserving this structure in paired comparisons makes the evidence more interpretable. Biological case studies can then link selected genes and pathways to the output through interventions, using the represented gene universe for annotation and enrichment calculations.

6 Discussion

6.1 Relational dependence across architectures

Biological priors act at different stages of a predictor: connectivity organizes context, auxiliary learning shapes initialization, and attention controls conditional readout. Parallel encoders add direct access to molecular measurements. Evaluating these routes together connects model design to the biological and predictive questions it is intended to answer.

The empirical contribution is the conjunction of three observations: B4's relational sensitivity, recurrent attention selections and context-varying conditional-omics attributions. Graph dependence and readout–attribution agreement describe distinct properties of this architecture. Their coexistence shows why a predictive result becomes more informative when it is connected to the route that carries molecular information into the output.

This perspective also locates biological interpretation at the interface between data assembly and computation. Restricting annotations to model genes changes which priors are represented for four additional drugs. Coverage tables and explicit explanation functions therefore contribute directly to the scientific account of a model. Together with the architecture-specific sensitivity profile, they provide a basis for selecting the comparisons that resolve a particular biological or predictive claim.

6.2 Scope and extensions

The empirical findings concern single historical runs within one GDGN data assembly and ten development-selected interpretation cases. The random-pair setting, conditional-omics attribution function and specified graph perturbations define their scope. A4/B4 compare complete architectures; A1/A4 compare composite initialization and representation settings.

Matched retraining extends the framework to topology benefit and transfer, while branch-conditioned attribution and interventions connect readout to output dependence. Preselected interpretation cases, repeated seeds and entity-held-out tasks extend its evaluation across contexts. Appendix A records the detailed provenance and annotation definitions needed to carry these comparisons forward.

7 Conclusion

Evaluating biological priors requires connecting a modelling claim to the information route and experimental contrast that reveal it. This review brings attention, transfer and topology into one diagnostic framework. The GDGN case study demonstrates its value through sharply architecture-dependent graph sensitivity, distinct readout and attribution profiles, and coverage that changes with the represented gene universe. These findings give a concrete account of how biological information participates in a predictive system and a focused basis for designing interpretable drug response models.

Data and code availability

The accompanying package contains LaTeX sources, verified references, vector figures, machine-readable summaries, source hashes and the reanalysis scripts. The case-study values derive from local GDGN records through B4; record-level provenance is documented in the package.

A Evidence definitions and record provenance

The review is a thematic narrative synthesis of primary methods, resources and evaluation studies. The empirical component uses existing GDGN configurations, test reports, perturbation summaries and interpretation records. All numerical calculations for the case study are deterministic reanalysis; the manuscript introduces no new training or checkpoint inference. Historical reports labelled LODO or LOCO grouped predictions from a pairwise-trained checkpoint over the response pool. Predictive comparisons here use the random-pair test reports; entity-held-out protocols appear as evaluation designs.

The ten interpretation cases were selected during development using relative sensitivity and resistance. Their 45 pairwise overlaps share cases. Saved JSON records supply attention–IG correlations and ranked genes; the conditional-omics function and L2 aggregation were recovered from the historical implementation. The records do not retain full attention vectors, checkpoint hashes, integration/reference settings or numerical completeness diagnostics.

PPI and DTI are operational labels. The supplied preprocessing scripts use STRING v12.0 associations with raw `combined_score > 700` and DGIdb drug–gene records with raw `interaction_score > 0.175`. DGIdb records include interaction annotations beyond direct binding. The local DGIdb release is unspecified; its resource citation describes database semantics. Coverage counts are reconstructed from supplied annotations, compound mappings and core genes, without loading original edge tensors. The six evaluable recovery cases have one to three represented drug-associated genes each; the other four are outside the annotation-recovery denominator. Annotation overlap, computational dependence and biological mechanism are separate interpretation outcomes.

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